

The exact bridge from cyclic interpolation volumes to the two Toda flows

13 September 2026

This is a standalone continuation of the identities CV.1–CV.12. It retains the original polynomial quotient, arithmetic measure, monic coordinates, arithmetic unit, source relations and complex phase. The delivered `Tau_Toda_Volume_Control/NOTE.tex` was read in full from its original archive; its SHA256 is

027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375.

The calculations below prove the exact maps underlying the determinant quotient, its first and second tilt derivatives, the omitted-direction determinant, the positive minor sum and the finite upper bound. No asymptotic arithmetic estimate is asserted.

1 Original arithmetic objects and the strip of real tilts

The original spaces and transforms are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for every } b \in \mathbb{Z}, j \geq 0\}, \\ D &= -x\partial_x, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{M}F(s) &= \int_0^\infty F(x) x^{s-1} dx. \end{aligned}$$

The series and its Euler derivatives converge uniformly at large x with every power of decay. For small x , Poisson summation with Fourier kernel $e^{-2\pi ixy}$ gives $\Theta\phi(x) = x^{-1}\Theta\hat{\phi}(1/x)$; both zero terms vanish because $\phi(0) = \hat{\phi}(0) = 0$. The Fourier transform preserves the stated two zero constraints and evenness. This formula proves every small- x seminorm after Euler differentiation, hence $\Theta : V \rightarrow \mathcal{B}$. It also commutes with D on these spaces by termwise differentiation. Mellin transforms of elements of $\mathcal{B}$ are entire, and integration by parts gives $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$; both endpoint terms vanish with their original coefficients.

Let h be a monic, nonempty packet of actual nontrivial zeros of

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with their complete orders. Assume the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Set $d = \deg h$ and retain

$$v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = [v_h]_h \in E_h^\times, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}. \quad (\text{TVB.1})$$

Here the jet of an entire function f means its unique polynomial Hermite remainder of degree less than d : in the local factor $\mathbb{C}[t]/(t^{m_\rho})$, it is $\sum_{j < m_\rho} f^{(j)}(\rho)t^j/j!$, with $t = s - \rho$. The coprime

factors of h give the isomorphism to their product by monic division and Bezout's identity, which supplies the unique global remainder. The constant $v_h(\rho)$ is nonzero at each selected zero because its full order was divided out. If the local jet is $\sum a_j t^j$, its inverse coefficients are $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$ for $n < m_\rho$. Multiplication verifies this inverse in every local factor, and the product isomorphism gives the inverse of v_h .

The original F_h has a construction within $\mathcal{B}$. For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy.$$

The equality is the vanishing Mellin value. For any desired power bound at infinity choose $A > \Re \rho$ with $|F(y)| \leq C_A y^{-A}$ for $y \geq 1$; the first integral is bounded by $C_A x^{-A}/(A - \Re \rho)$ there. At zero choose B with $B + \Re \rho > 0$ and $|F(y)| \leq C_B y^B$ for $0 < y \leq 1$; the second integral is bounded by $C_B x^B/(B + \Re \rho)$. The exact identity $(D - \rho)S_\rho F = F$ now gives all Euler derivative bounds inductively, so $S_\rho F \in \mathcal{B}$. Mellin integration by parts proves $\mathcal{M}(S_\rho F) = \mathcal{M}F/(s - \rho)$, with its removable value at $s = \rho$ supplied by the derivative of the numerator. For the original seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V$, direct substitution $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad \Re s > -2.$$

For $\Re s > 1$, absolute convergence permits the two-sided theta sum to be integrated termwise, yielding $\mathcal{M}(\Theta\phi_0) = 2\zeta(s)\mathcal{M}\phi_0(s) = g(s)$; both sides continue to the same entire function. Apply S_ρ successively to $\Theta\phi_0$ over the full list of roots of h , including each repeated occurrence. At every stage the next vanishing Mellin value holds by its remaining order. This constructs $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$ and $h(D)F_h = \Theta\phi_0$; the latter also follows by Mellin injectivity, which is Fourier injectivity after $x = e^y$. Fix $k \geq 1$, set $c = k/2$, $S = s_1 + \dots + s_k$, and put $I = (h(s_1), \dots, h(s_k)) \subset \mathbb{C}[s_1, \dots, s_k]$. Let χ be the original monic generator of $\{P \in \mathbb{C}[S] : P(S) \in I\}$. Its degree is $q \geq 1$ and

$$C = \mathbb{C}[S]/(\chi), \quad A = M_S : C \longrightarrow C. \quad (\text{TVB.2})$$

The coordinate basis of C is always $1, S, \dots, S^{q-1}$. The polynomial exists because multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$ has an annihilating polynomial. Division by its least-degree monic annihilator proves that its entire annihilating ideal is (χ) . In particular the injection

$$\eta : C \longrightarrow (E_h^{\otimes k})^{\mathfrak{S}_k}, \quad [P] \longmapsto v_h^{\otimes k}[P(S)]_I \quad (\text{TVB.3})$$

has exactly zero kernel: multiplication by the displayed unit is invertible and the preceding annihilator ideal describes the other factor's kernel. The original tensor source map is

$$\mathcal{V}P = P(D_1 + \dots + D_k)F_h^{\otimes k}, \quad D = -x\partial_x, \quad \mathcal{M}F_h = v_h.$$

On the original Mellin line its scalar observation has squared norm

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad \int m_k = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{TVB.4})$$

Indeed the tensor Mellin–Plancherel density is $\prod_i w_h(t_i) d^k t$. In coordinates $u = \sum_i t_i$, $y = (t_1, \dots, t_{k-1})$, the inverse is $(y, u) \mapsto (y_1, \dots, y_{k-1}, u - \sum y_i)$. With (y, u) in this order its determinant is 1; with (u, y) its determinant is $(-1)^{k-1}$. Integrating the original density over y gives (TVB.4), with no scalar divisor.

For every compact substrip $|\Re \theta| \leq r < \pi/2$, all integrals

$$M_h(\theta) = \int e^{\theta t} w_h(t) dt, \quad Z(\theta) = M_h(\theta)^k = \int e^{\theta u} m_k(u) du \quad (\text{TVB.5})$$

and all their polynomially weighted derivatives converge locally uniformly. Here is an elementary sufficient bound. Summation by parts for the alternating Dirichlet series, whose partial sums

have modulus at most one, gives $|\eta(s)| \leq |s|/\Re s$ when $\Re s > 0$. On $\Re s = 1/2$, the denominator in $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ has modulus at least $\sqrt{2} - 1$. For $0 < \varphi < \pi/2$, rotate the defining gamma integral to the ray with argument $\text{sgn}(t)\varphi$. The large arc vanishes by $e^{-R\cos\varphi}$ and the small arc by the positive real part $1/4$ of its exponent. Its absolute value is bounded by

$$|\Gamma(1/4 + it/2)| \leq \Gamma(1/4)(\cos\varphi)^{-1/4}e^{-\varphi|t|/2}.$$

For sufficiently large $|t|$, the monic polynomial h satisfies $|h(1/2 + it)| \geq C_h(1 + |t|)^d$ with $C_h > 0$. The two factors $s, s - 1$, the preceding zeta bound and the gamma bound consequently give

$$w_h(t) \leq C_{h,\varphi}(1 + |t|)^{6-2d}e^{-\varphi|t|}$$

outside a compact interval. On that interval v_h is entire, since the selected orders are complete. Choose $r < \varphi < \pi/2$ and absorb every fixed polynomial in the remaining exponential decay. This proves the asserted domination and differentiation under the integral. Fubini gives the exact product M_h^k in (TVB.5). The nonzero entire function v_h has a discrete real-line zero set; therefore $w_h > 0$ almost everywhere. Its convolutions are positive almost everywhere (in fact at every u for $k \geq 2$), since their fibre integrands are positive off a finite union of measure-zero inverse images of this discrete set. Thus the norm of every nonzero polynomial is positive.

In what follows θ is real, $|\theta| < \pi/2$, and the source pairing, conjugate-linear in its first argument, is exactly

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (\text{TVB.6})$$

The map $f(u) \mapsto e^{\theta u/2} f(u)$ is an onto isometry from $L^2(e^{\theta u} m_k du)$ to $L^2(m_k du)$, with inverse $f \mapsto e^{-\theta u/2} f$. It commutes with multiplication by $c + iu$. This specifies the exact Hilbert comparison for the tilt. The polynomial algebra, its reduction and η are unchanged. When the target of $\mathcal{V}$ is observed at this tilt, its declared tensor norm is

$$\|F\|_{k,\theta}^2 = \int_{\mathbb{R}^k} e^{\theta \sum_i t_i} |\mathcal{M}_{\text{unit}} F(t_1, \dots, t_k)|^2 d^k t,$$

where $\mathcal{M}_{\text{unit}}$ is the original Mellin–Plancherel map with a factor $(2\pi)^{-1/2}$ in each coordinate. The map $f(t) \mapsto e^{\theta \sum_i t_i/2} f(t)$ is an onto isometry from this weighted Mellin Hilbert space to the original one, with the opposite exponential as inverse. The restriction to $\mathcal{V}P$ is exactly (TVB.6), by the same sum-fibre integral as (TVB.4). Thus every tilted exterior norm below has a specified target norm; at $\theta = 0$ it is the original tensor-source norm.

2 The source relation and its Laplace observation

Write $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = 0$. For $N \geq q - 1$, put $n = N + 1$, $m = N - q + 1$. Literal monic division gives the exact sequence

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{T_N} C \longrightarrow 0, \quad M_\chi Q = \chi Q, \quad T_N P = [P]_\chi. \quad (\text{TVB.7})$$

Its kernel is precisely the indicated image, and its first q monomial columns prove surjectivity. Denote that kernel by $\mathcal{D}_N = \chi \mathcal{P}_{m-1}$.

This sequence has an explicit map to the original theta boundaries. To see every sign, choose the quotients Q_i from successive monic division, in the order $s_1, \dots, s_k$, so that

$$\chi(S) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{TVB.8})$$

The remainder is zero by the definition of χ and uniqueness of the tensor remainder basis. Put $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$. It is even, vanishes at zero and has integral zero. Direct gamma

integration of its two terms gives $\mathcal{M}(\Theta\phi_0) = g$, hence $h(D)F_h = \Theta\phi_0$ by Mellin injectivity on the original rapidly decreasing source. For every polynomial P , an actual primitive of $\mathcal{V}(\chi P)$ in the tensor complex $(V \xrightarrow{\Theta} \mathcal{B})^{\otimes k}$ is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{TVB.9})$$

The preceding $i-1$ factors have cochain degree one. The tensor differential therefore contributes $(-1)^{i-1}$ at the i th factor, cancelling the written sign and yielding (TVB.8). Euler differentiation preserves the source conditions and commutes with Θ , so all terms have the indicated domains. Conversely the full jet of $\mathcal{V}P$ is $\eta T_N P$; injectivity in (TVB.3) shows that the admitted polynomial source has no further zero-jet elements. In particular every boundary vector in (TVB.7) retains its positive source norm before the quotient sends it to represented zero.

For any polynomial P , define its Laplace observation by

$$\mathcal{L}(P)(\theta) = \int P(c + iu) e^{\theta u} m_k(u) du.$$

Differentiation under (TVB.5) proves the typed intertwining $\mathcal{L}(SP) = (c + i\partial_\theta)\mathcal{L}(P)$. Coefficientwise conjugation gives the exact Gram observation

$$Z_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta)Z = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0. \quad (\text{TVB.10})$$

Thus the two signs and both polynomial factors arise from $\langle M_\chi 1, M_\chi 1 \rangle_\theta$. This is a map on source functions, with no replacement of the higher contracted ideal by a power of χ .

3 Both monic determinants and their exact determinant-line map

Set

$$\mathfrak{D}_j = \det[Z^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{B}_j = \det[Z_\chi^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{D}_0 = \mathfrak{B}_0 = 1. \quad (\text{TVB.11})$$

All nonempty determinants are positive by polynomial positivity. The coefficient map from $1, S, \dots, S^{j-1}$ to $1, u, \dots, u^{j-1}$ is explicitly

$$S^a = (c + iu)^a = \sum_{b=0}^a \binom{a}{b} c^{a-b} i^b u^b. \quad (\text{TVB.12})$$

Its determinant is $i^{j(j-1)/2}$ and its Gram determinant multiplier is exactly one. Therefore $\mathfrak{D}_j$ is also the original S -monomial Gram determinant. The same calculation applies to the relation-weight pairing $\langle P, Q \rangle_{\chi, \theta} = \langle \chi P, \chi Q \rangle_\theta$ and $\mathfrak{B}_j$. This proves the coordinate correspondence without changing the original monic polynomial.

Let $p_j(S; \theta)$ be the monic source orthogonal polynomials, and $r_j(S; \theta)$ the monic polynomials for this relation-weight pairing. Retain their original squared norms

$$\omega_j = \|p_j\|_\theta^2 = \mathfrak{D}_{j+1}/\mathfrak{D}_j, \quad \nu_j = \|\chi r_j\|_\theta^2 = \mathfrak{B}_{j+1}/\mathfrak{B}_j. \quad (\text{TVB.13})$$

The equations follow by triangular monic Gram elimination, whose diagonal change is one. In particular $\omega_0 = M_h(\theta)^k$. The full mass has not been divided out.

Give C its written coordinate Euclidean pairing only when specifying adjoints. Let M_N be the original monomial source Gram, and J_N its reduction matrix. The minimum lift is

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = M_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N. \quad (\text{TVB.14})$$

Here the last expression specifies polynomial coefficients. It satisfies $T_N R_N = I$, $\text{im } R_N \perp \mathcal{D}_N$ and $R_N^* R_N = G_N$ in the original source pairing. Indeed the first equality is matrix multiplication, and pairing with any vector in $\ker J_N$ gives zero. Every other lift is $R_N x + B$ for $B \in \mathcal{D}_N$, so its squared norm is $x^* G_N x + \|B\|^2$. This proves the minimum assertion and uniqueness.

Use the source basis $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$. Its change from the n source monomials is triangular with diagonal one. Replacing its first q columns by $R_N[1], \dots, R_N[S^{q-1}]$ adds boundary columns and again has determinant one. The resulting Gram is the direct sum of G_N and the relation monomial Gram. Consequently

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_m}, \quad D_N^{\text{CV}} := \det K_N = V_N^{-1}. \quad (\text{TVB.15})$$

More precisely the map $\det C \otimes \det \mathcal{P}_{m-1} \rightarrow \det \mathcal{P}_N$ sends the written coordinate wedges to the wedge of these quotient lifts followed by M_χ of the relation monomials; its coordinate determinant is 1. Putting the kernel wedge first instead contributes the explicit sign $(-1)^{qm}$. Its norm identity is $\det G_N \mathfrak{B}_m = \mathfrak{D}_n$. Composing the wedge with $\mathcal{V}^{\otimes n} \text{Alt}_n$ multiplies its squared norm by exactly $n!$: expanding the unscaled alternating tensor sum gives $n!$ copies of the Gram determinant. The original tensor target is $\mathcal{B}^{\otimes kn}$ in cochain degree kn .

4 Toda derivatives and the full curvature operators

For either positive real u -weight in (TVB.11), write $Q_j(u)$ for its real monic orthogonal polynomials and h_j for their norms. Orthogonality and leading coefficients prove

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1} \ (j \geq 1), \quad a_0 = 0.$$

For $i \leq j-2$, differentiating $\langle Q_i, Q_j \rangle = 0$ shows that $\partial_\theta Q_j$ is orthogonal to Q_i , because uQ_i and $\partial_\theta Q_i$ have degree at most $j-1$. For $i = j-1$ the same identity gives its coefficient $-a_j$. Thus $\partial_\theta Q_j = -a_j Q_{j-1}$ and $h'_j = b_j h_j$. To compute b'_j , differentiate $h_j b_j = \langle Q_j, uQ_j \rangle$. The two polynomial derivative terms contribute $-2a_j h_j$. The derivative of the measure contributes $\|uQ_j\|^2 = h_j(a_{j+1} + b_j^2 + a_j)$. After subtracting $h'_j b_j$ this gives $b'_j = a_{j+1} - a_j$. The monic Gram determinant is $\prod_{j < n} h_j$, hence

$$(\log \mathfrak{D}_n)'' = \frac{\omega_n}{\omega_{n-1}} \quad (n \geq 1), \quad (\log \mathfrak{B}_m)'' = \frac{\nu_m}{\nu_{m-1}} \quad (m \geq 1). \quad (\text{TVB.16})$$

Both 0th logarithms have second derivative zero. Multiplication by the square of the determinant proves each Hankel–Toda equation with coefficient one. Set

$$\alpha_{N+1} = \omega_{N+1}/\omega_N, \quad \beta_m = \nu_m/\nu_{m-1} \ (m \geq 1), \quad \beta_0 = 0, \quad \ell_N = \log V_N.$$

Then $\ell_N'' = \alpha_{N+1} - \beta_m$.

There is a finer exact source-map identity. Let $X = (S - c)/i$, which is multiplication by the real variable u in (TVB.6). Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ to $\mathcal{D}_N$ in that pairing. Differentiating $T_N R_N = I$ gives $R'_N x \in \mathcal{D}_N$. Differentiating orthogonality against each fixed boundary polynomial gives

$$R'_N = -P_{\mathcal{D}_N} X R_N =: -I_N, \quad O_N := (1 - P_N) X R_N. \quad (\text{TVB.17})$$

Every derivative is an actual boundary with the primitive (TVB.9). Differentiating its norm, including the derivative of the measure, gives

$$G'_N = R_N^* X R_N, \quad G''_N = R_N^* X^2 R_N - 2R_N^* X P_{\mathcal{D}_N} X R_N.$$

The projection to $\text{im } R_N$ is $R_N G_N^{-1} R_N^*$ and, together with $P_{\mathcal{D}_N}$, sums to P_N . Subtracting the square of the first derivative therefore proves

$$G''_N - G'_N G_N^{-1} G'_N = O_N^* O_N - I_N^* I_N. \quad (\text{TVB.18})$$

This is an equality of maps $C \rightarrow C^{\text{anti}*}$, where $C^{\text{anti}*}$ consists of conjugate-linear functionals on C . For example $O_N^* O_N$ sends x to the functional $y \mapsto \langle O_N y, O_N x \rangle_\theta$; this assignment is complex-linear in x . The fixed coordinate Euclidean pairing identifies that antidual with the written matrix coordinates.

Write $b_j = [p_j]_\chi$, and $a = b_N^* G_N b_N$, $d_+ = b_{N+1}^* G_N b_{N+1}$. The canonical next boundary has the particularly exact factorization

$$\mathbf{b}_N := p_{N+1} - R_N b_{N+1} = \chi r_m, \quad \nu_m = \omega_{N+1} + d_+. \quad (\text{TVB.19})$$

To prove it, χr_m is monic of degree $N+1$ and orthogonal to $\mathcal{D}_N$. Subtract p_{N+1} ; the difference lies in $\mathcal{P}_N \cap \mathcal{D}_N^\perp$ and has reduction $-b_{N+1}$. Uniqueness of R_N proves the first equality; the two terms in its definition are orthogonal, giving the second. The coefficient of p_N in $R_N x$ is $b_N^* G_N x / \omega_N$. The leading coefficient of X as an S -polynomial is $1/i = -i$. Consequently the full maps are

$$O_N x = -i p_{N+1} \frac{b_N^* G_N x}{\omega_N}, \quad I_N x = -i \chi r_{m-1} \frac{b_{N+1}^* G_N x}{\nu_{m-1}} \quad (m \geq 1), \quad I_N = 0 \quad (m = 0). \quad (\text{TVB.20})$$

For the second equation, $X \chi r_j$ is an old boundary if $j \leq m-2$. For $j = m-1$ its coefficient of χr_m is $-i$. Moving X across the pairing conjugates this coefficient to i , and $\langle \chi r_m, R_N x \rangle = -b_{N+1}^* G_N x$ by (TVB.19). Division by the original norm ν_{m-1} gives exactly the second sign in (TVB.20).

Both maps have rank at most one. For $N \geq q$, put $\delta_N = V_N / V_{N-1}$. The rank-one calculation in the next section gives $a/\omega_N = 1 - \delta_N$ and $\delta_N = \omega_N / \nu_{m-1}$. Taking traces in (TVB.20) gives the separate, positive source costs

$$\begin{aligned} \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1} (1 - \delta_N), \\ \text{Tr}(G_N^{-1} I_N^* I_N) &= \beta_m - \alpha_{N+1} \delta_N = \frac{\nu_m - \omega_{N+1}}{\nu_{m-1}}. \end{aligned} \quad (\text{TVB.21})$$

Their difference is $\alpha_{N+1} - \beta_m$ as required. Thus the individual positive map costs each retain the common quantity $\alpha_{N+1} \delta_N$; assigning them separately the two Toda coefficients would lose this quantity. At $N = q-1$, the outer cost is α_q and the inner cost is zero. If a single notation for the cost identities is wanted at this endpoint, define only the scalar convention $\delta_{q-1}^{\text{end}} := \det K_{q-2} / \det K_{q-1} = 0$; the included column has leverage $a/\omega_{q-1} = 1$. Together with $\beta_0 = 0$ the cost identities are $\alpha_q (1 - \delta_{q-1}^{\text{end}}) = \alpha_q$ and $\beta_0 - \alpha_q \delta_{q-1}^{\text{end}} = 0$. This convention is not a quotient involving V_{q-2} and does not introduce an inverse for K_{q-2} or a norm ν_{-1} . All statements concern the original polynomial source and its original boundary subspace.

5 The rank-one update, trace phase and original rank-two radius

The source-orthogonal basis gives

$$K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad K_{N+1} = K_N + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}. \quad (\text{TVB.22})$$

The first q monic remainder columns are a triangular basis; K_N is positive definite precisely in the admitted range. Block elimination of $\begin{pmatrix} K & U \\ -V^* & I \end{pmatrix}$ in either order proves $\det(K + UV^*) = \det K \det(I + V^* K^{-1} U)$. Applying it to the positive update and the subtraction of b_N from K_N yields, for $N \geq q$,

$$\frac{a}{\omega_N} = 1 - \delta_N, \quad \frac{d_+}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1, \quad \delta_N = \frac{\omega_N}{\nu_{m-1}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_m}. \quad (\text{TVB.23})$$

The last two identities also follow directly from (TVB.15). Each δ is positive and at most one, since increasing the source space weakly decreases every minimum squared norm. The exact inverse update is

$$G_{N+1} = G_N - \frac{G_N b_{N+1} b_{N+1}^* G_N}{\omega_{N+1} + d_+}. \quad (\text{TVB.24})$$

Multiplying this expression by K_{N+1} proves it directly; the denominator is $\nu_m > 0$ by (TVB.19).

Reflection of the tensor algebra sends S to $k - \bar{S}$ and preserves its annihilator ideal. Monicity gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic. The exact algebra isomorphism

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P(S)] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q(u)] = [Q((S - c)/i)] \quad (\text{TVB.25})$$

transports the original ideal by its leading unit i^q . Its coefficient matrix E is (TVB.12) in dimension q . Thus A transports to $cI + iT$, where $T = M_u$ in the real quotient. Its Gram transports to $E^{-*}G_N E^{-1}$, its kernel to $E K_N E^*$, and $\det G_N$ is unchanged because $|\det E| = 1$. Write

$$\sigma = \text{Tr } T = \frac{\text{Tr } A - cq}{i} \in \mathbb{R}, \quad \psi_N = \sigma - \ell'_N.$$

Reality of $\hat{\chi}$, the weight and its monic orthogonal polynomials makes all the u -coordinate source and quotient Grams real. Let $Q_j(u) = i^{-j} p_j(c + iu)$ and $d_j = [Q_j]$. Then $E b_j = i^j d_j$ and both source norms are ω_j .

The leading coefficient of the u -polynomial minimum lift of x is $d_N^* G_N x / \omega_N$, where in this calculation G_N denotes the transported real-coordinate Gram. The difference $X R_N - R_N T$ has zero reduction, and subtracting $(Q_{N+1} - R_N d_{N+1}) d_N^* G_N / \omega_N$ cancels its highest coefficient. Its remainder is an old boundary. Pair with R_N ; old boundaries vanish and $R_N^* (Q_{N+1} - R_N d_{N+1}) = -G_N d_{N+1}$. Taking the trace after multiplication by G_N^{-1} proves

$$\ell'_N = \sigma - \frac{d_N^* G_N d_{N+1}}{\omega_N}, \quad \frac{b_N^* G_N b_{N+1}}{\omega_N} = i \psi_N. \quad (\text{TVB.26})$$

The second equation is back in the fixed original S coordinates: the phase is $\overline{i^N} i^{N+1} = i$. Both signs are retained.

For completeness, the u recurrence proved above transports to $S p_j = p_{j+1} + (c + i b_j^{\text{rec}}) p_j - \alpha_j p_{j-1}$ in the original source coordinates. Inserting this in (TVB.22) telescopes all interior terms and gives

$$A K_N + K_N A^* - k K_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N}.$$

For $W_N = A^* G_N + G_N A - k G_N$, multiplication on both sides by G_N gives the corresponding identity for W_N . Its Hermitian relative operator is $H_N = G_N^{-1/2} W_N G_N^{-1/2}$. It has rank at most two and trace $2\Re \text{Tr } A - kq = 0$. The sum of the squares of its eigenvalues is $2(ad_+ - |z_N|^2)/\omega_N^2$, where $z_N = b_N^* G_N b_{N+1} = i \omega_N \psi_N$. This last assertion follows by expanding the square of $(xy^* + yx^*)/\omega_N$ for $x = G_N^{1/2} b_N$, $y = G_N^{1/2} b_{N+1}$; its trace is $[2ad_+ + 2\Re(z_N^2)]/\omega_N^2$. Its nonzero eigenvalues are therefore $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2. \quad (\text{TVB.27})$$

The first expression is valid already at $N = q - 1$.

6 Exact maps to the CV omitted determinant and positive minors

Retain $D_j^{\text{CV}} = \det K_j$, put $K_{-1} = 0$, and form

$$\tilde{K}_N = K_{N-1} + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}, \quad \tilde{D}_N = \det \tilde{K}_N, \quad \Delta_N = D_{N+1}^{\text{CV}} + D_{N-1}^{\text{CV}} - D_N^{\text{CV}} - \tilde{D}_N.$$

This is the actual quotient kernel of the source with p_N omitted and p_{N+1} included. No inverse of it is used. The two-column determinant formula preceding (TVB.23), with one negative and one positive update, proves

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} = \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d_+}{\omega_{N+1}}\right) + \frac{|z_N|^2}{\omega_N \omega_{N+1}}, \quad \epsilon_N^2 = \alpha_{N+1} \frac{\Delta_N}{D_N^{\text{CV}}}. \quad (\text{TVB.28})$$

In particular, for $N \geq q$, the complete phase map is the exact nonnegative difference

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} - \frac{\delta_N}{\delta_{N+1}} = \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.29})$$

The sign of ψ_N is retained by the first derivative in (TVB.26); the omitted determinant observes its square.

For any finite column set L , determinant multilinearity gives

$$\det \left(\sum_{j \in L} \frac{b_j b_j^*}{\omega_j} \right) = \sum_{\substack{J \subseteq L \\ |J|=q}} \frac{|\det B_J|^2}{\prod_{j \in J} \omega_j}, \quad B_J = [b_j]_{j \in J \text{ increasing}}.$$

Terms with repeated indices vanish. The $q!$ remaining permutations of a fixed increasing index set yield its determinant times its conjugate, which proves the displayed identity including singular kernels. Apply it to the four column sets defining Δ_N . A set containing neither of the two last indices has coefficient $1 + 1 - 1 - 1 = 0$; one containing exactly one has coefficient $1 - 1 = 0$; one containing both has coefficient one. Hence

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \quad (\text{TVB.30})$$

For $q = 1$ it is zero. This proof applies at $N = q - 1$. For $q \geq 2$ define the actual source subspace $\mathcal{E}_N = (\bigwedge^{q-2} \mathcal{P}_{N-1}) \wedge p_N \wedge p_{N+1}$ and $\Lambda_N = (\bigwedge^q T_{N+1})|_{\mathcal{E}_N}$. For the fixed coordinate wedge $e_C = 1 \wedge S \wedge \dots \wedge S^{q-1}$, whose coordinate Euclidean squared norm is one, the adjoint is

$$\zeta_N := \Lambda_N^* e_C = \sum_{|I|=q-2} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \left(\bigwedge_{i \in I} p_i \right) \wedge p_N \wedge p_{N+1}. \quad (\text{TVB.31})$$

The displayed source wedges are orthogonal with the written full product norms. Therefore $\|\zeta_N\|_\theta^2 = \Delta_N$ and $\Lambda_N \zeta_N = \Delta_N e_C$. At $\theta = 0$ its literal original tensor-source image has squared norm $q! \Delta_N$ under $\mathcal{V}^{\otimes q} \text{Alt}_q$. At other real tilts use the onto Hilbert comparison after (TVB.6) on each scalar source factor; the same determinant norm and factorial hold in that specified tilted observation. For $q = 1$ set $\mathcal{E}_N = 0$, $\Lambda_N = 0$, $\zeta_N = 0$.

7 Both exact losses, their equality cases and the telescope

For $N \geq q$ define

$$v_N = -\log \delta_N, \quad s_N = (v_N + v_{N+1})/2, \quad t_N = (v_{N+1} - v_N)/2, \quad \mathcal{R}_N = e^{2s_N} = \frac{V_{N-1}}{V_{N+1}} \geq 1.$$

Expanding $(1 - e^{-v_N})(e^{v_{N+1}} - 1)$ gives $2e^{t_N} \cosh s_N - e^{2t_N} - 1$. Completing this scalar square, with both terms retained, yields

$$\boxed{\epsilon_N^2 = \alpha_{N+1} \sinh^2 s_N - \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 - \psi_N^2} \quad (\text{TVB.32})$$

Thus the exact upper bound and its equality conditions are

$$\epsilon_N \leq \sqrt{\alpha_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}, \quad \text{equality} \iff \psi_N = 0 \text{ and } \delta_N(2 - \delta_{N+1}) = 1. \quad (\text{TVB.33})$$

Indeed $\alpha_{N+1} > 0$ and both lost squares in (TVB.32) are nonnegative. Multiplication of $e^{t_N} = \cosh s_N$ by $2\sqrt{\delta_N \delta_{N+1}}$ gives precisely the displayed equation. This includes the case $\mathcal{R}_N = 1$, when both contractions equal one and $\epsilon_N = \psi_N = 0$ by (TVB.27). The corresponding equality with the full positive minor source is

$$\frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N} - \frac{\|\zeta_N\|_\theta^2}{D_N^{\text{CV}}} = (e^{t_N} - \cosh s_N)^2 + \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.34})$$

It follows from (TVB.28), (TVB.31) and (TVB.32), so its minor equality has the same two necessary and sufficient conditions. In particular equality is a fully specified source-volume equation, not a consequence of positivity alone.

There is a direct formulation in the actual volumes, without logarithmic variables. Put $U_N = \sqrt{\alpha_{N+1}}(\mathcal{R}_N - 1)/(2\sqrt{\mathcal{R}_N})$. Since

$$e^{t_N} = \frac{V_N}{\sqrt{V_{N-1}V_{N+1}}}, \quad \cosh s_N = \frac{V_{N-1} + V_{N+1}}{2\sqrt{V_{N-1}V_{N+1}}},$$

equation (TVB.32) gives the exact two losses

$$U_N^2 - \epsilon_N^2 = \alpha_{N+1} \frac{(2V_N - V_{N-1} - V_{N+1})^2}{4V_{N-1}V_{N+1}} + \frac{|z_N|^2}{\omega_N^2}. \quad (\text{TVB.34a})$$

Thus the volume equality is the arithmetic midpoint $2V_N = V_{N-1} + V_{N+1}$ together with $z_N = 0$. Equal consecutive logarithmic contractions alone do not give this midpoint: their extra equation $V_N^2 = V_{N-1}V_{N+1}$ implies it only if all three positive volumes are equal, by the square $(\sqrt{V_{N-1}} - \sqrt{V_{N+1}})^2$. In the original inverse-volume coordinates $D_- = D_{N-1}^{\text{CV}}$, $D_0 = D_N^{\text{CV}}$, $D_+ = D_{N+1}^{\text{CV}}$ and $E_N = \tilde{D}_N$, the same identities become

$$\begin{aligned} E_N D_0 - D_- D_+ &= \frac{D_0^2 |z_N|^2}{\omega_N \omega_{N+1}}, \\ \frac{D_0(D_+ - D_-)^2}{4D_- D_+} - \Delta_N &= \frac{(2D_- D_+ - D_0(D_- + D_+))^2}{4D_0 D_- D_+} + E_N - \frac{D_- D_+}{D_0}. \end{aligned} \quad (\text{TVB.34b})$$

Substitute $D_j = V_j^{-1}$ in (TVB.34a), or multiply out its two squares, to verify every denominator. Both terms on the right are nonnegative by the first equation. Equality is equivalently $D_0 = 2D_- D_+ / (D_- + D_+)$ and $E_N = (D_- + D_+)/2$; it then gives $\Delta_N = E_N - D_0$. In particular phase zero at these degrees records a positive omitted-direction determinant $E_N = D_- D_+ / D_0$, with its exact map specified in (TVB.28), rather than a zero determinant.

The algebraic exterior lower bound applies at each real tilt as well as at zero. Here is a direct proof. In the Hermitian matrix H_N above let P be the orthogonal projection to the image, under $G_N^{1/2}$, of the sum of all full generalized eigenspaces of A with $\Re \lambda > c$. Invariance of that subspace makes the transformed A upper triangular in its orthogonal direct sum with the complement, without asserting that the spectral projection itself is orthogonal. Taking the trace of its first diagonal block gives

$$\text{Tr}(PH_N) = L := \sum_{\Re \lambda > c} \ell_\lambda (2\Re \lambda - k).$$

Since H_N has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros, $\text{Tr}(PH_N) \leq \epsilon_N$: its positive eigenline contributes at most ϵ_N and its negative eigenline contributes a nonpositive number. All algebraic multiplicities

ℓ_λ are retained. Thus $L \leq \epsilon_N$ for the same tilted minimum metric, without choosing another quotient metric.

Combining this with (TVB.33), using monotonicity of $\sinh$ on $[0, \infty)$ and its inverse, proves

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.35})$$

For $N = N_0 + 2j$, $0 \leq j < r$, with $N_0 \geq q$, every denominator in one logarithm is the numerator in the next. Summing the exact logs and then their lower bounds proves

$$\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N_0+2j+1}}} \right), \quad r \geq 1. \quad (\text{TVB.36})$$

Every term is at the same fixed real tilt, packet and tensor degree. No relation between different k or a limiting value of either determinant is used. Under $\mathcal{V}$ the minimum representatives at all degrees retain jets η and their original quotient classes; each difference of two such lifts is the explicit boundary in (TVB.7)–(TVB.9).

8 The first admitted degree and the empty quotient

At $N = q - 1$, the first q monic remainder columns are a basis, so $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $i, j < q$. Expand the original polynomial, with its coefficients retained,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle_\theta}{\omega_j}. \quad (\text{TVB.37})$$

Reduction gives $b_q = -\sum_{j<q} \gamma_j b_j$. Hence

$$a = \omega_{q-1}, \quad z_{q-1} = -\gamma_{q-1} \omega_{q-1}, \quad d_+ = \sum_{j<q} |\gamma_j|^2 \omega_j = \nu_0 - \omega_q, \quad \gamma_{q-1} = -i\psi_{q-1}.$$

The last equality follows from (TVB.26), which does not require K_{q-2}^{-1} . Substitution gives the exact CV.12–Toda endpoint

$$\epsilon_{q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - \psi_{q-1}^2. \quad (\text{TVB.38})$$

There is no V_{q-2} here. The omitted determinant instead records $\tilde{D}_{q-1}/D_{q-1}^{\text{CV}} = \psi_{q-1}^2/\alpha_q$. The polynomial projection $\mathbf{P}_{q-2}\chi = \sum_{j<q-1} \gamma_j p_j$ has the actual jet $\eta \sum_{j<q-1} \gamma_j b_j$, whereas χ itself has zero jet and primitive (TVB.9). Its squared source norm in (TVB.6) is the numerator in (TVB.38). This is the exact map from the whole relation to its lower orthogonal projection, including the possibly nonzero resulting observation. For $q = 1$, the sum is empty, the lower source is $\mathcal{P}_{-1} = 0$, and $\epsilon_N = 0$ at every admitted degree because its one-dimensional relative control is trace zero.

The endpoint projection and the minor vector have a stronger exact typed correspondence. Assume $q \geq 2$, let $D = D_{q-1}^{\text{CV}} = \prod_{j<q} \omega_j^{-1}$, and set $I_j = \{0, \dots, q-2\} \setminus \{j\}$ for $0 \leq j \leq q-2$. The determinant of the first q monic remainder columns is 1. Substituting $b_q = -\sum_{l<q} \gamma_l b_l$ in the minor with columns $I_j \cup \{q-1, q\}$ leaves only $l = j$. Moving b_j from its last position to position j takes $q-1-j$ transpositions; with the preceding minus sign this proves

$$\det B_{I_j \cup \{q-1, q\}} = (-1)^{q-j} \gamma_j. \quad (\text{TVB.38a})$$

Write $e_j = (\bigwedge_{i \in I_j} p_i) \wedge p_{q-1} \wedge p_q$. The explicit conjugate-linear isomorphism

$$\mathcal{H} : \mathcal{P}_{q-2} \longrightarrow \mathcal{E}_{q-1}, \quad \mathcal{H} \left(\sum_{j < q-1} a_j p_j \right) = \sum_{j < q-1} (-1)^{q-j} \overline{a_j} \frac{\omega_j D}{\omega_q} e_j \quad (\text{TVB.38b})$$

has inverse $\sum t_j e_j \mapsto \sum (-1)^{q-j} \omega_q \overline{t_j} p_j / (\omega_j D)$. The source and target basis vectors are orthogonal and $\|e_j\|_\theta^2 = \omega_q / (D\omega_j)$. Consequently

$$\|\mathcal{H}P\|_\theta^2 = \frac{D}{\omega_q} \|P\|_\theta^2, \quad \zeta_{q-1} = \mathcal{H}(P_{q-2}\chi), \quad \Delta_{q-1} = \frac{D}{\omega_q} \|P_{q-2}\chi\|_\theta^2. \quad (\text{TVB.38c})$$

This follows term by term from (TVB.31) and (TVB.38a). The actual alternating tensor image has squared norm $q!D\|P_{q-2}\chi\|_\theta^2/\omega_q$ in the specified tilted tensor observation, and in the original source norm at $\theta = 0$. The top coefficient is retained separately: $\tilde{D}_{q-1} = D|\gamma_{q-1}|^2\omega_{q-1}/\omega_q$. For $q = 1$ the map $\mathcal{H}$ is the unique map between zero spaces, with the same zero identities; the top-coefficient formula still records the entire omitted determinant.

For completeness the finite upper bound is strict when $q = 1$ and $N \geq 1$. Consecutive source polynomials have no common root: their three-term recurrence with its nonzero coefficient $\alpha_j > 0$ would propagate such a root backwards to $p_0 = 1$. For any nonzero quotient $q \geq 1$, both b_N, b_{N+1} therefore cannot vanish, since a root of χ would then be common to both polynomials. When $N \geq q$, K_{N-1} is positive definite and $K_{N+1} - K_{N-1}$ is a nonzero positive semidefinite matrix. Conjugating by $K_{N-1}^{-1/2}$ and multiplying its eigenvalues proves $D_{N+1}^{\text{CV}} > D_{N-1}^{\text{CV}}$, hence $\mathcal{R}_N > 1$ and $U_N > 0$. At $q = 1$ this is strictly larger than the already proved $\epsilon_N = 0$.

Finally let $h = 1$, so $\chi = 1$, $q = 0$, and $C = 0$. The analytic function g , w_1 and all source norms still exist. For every $N \geq 0$ the source sequence is multiplication by one from $\mathcal{P}_N$ onto itself followed by the zero quotient. Then $m = N + 1$, $Z_\chi = Z$, $\mathfrak{B}_m = \mathfrak{D}_m$, and the determinant of the zero-dimensional Gram is the empty determinant $V_N = 1$. Its logarithm and derivatives are zero. The minimum lift, its curvature maps, and the relative control are the unique zero maps, with $\epsilon_N = L = 0$. The two Toda coefficients in their curvature difference are equal, so that difference is zero. No positive-dimensional inverse is introduced. The support lift retains $G(0) = \{\tau, e\}$: every represented polynomial, including each nonzero source relation, maps to e , while τ maps to τ .

9 The complete join with the original spectral coupling

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Its two inputs are the delivered Toda equation (28), proved here as (TVB.32), and the complete original-metric exterior identity E8. The latter source is retained in the cyclic companion at commit 161089942cce70a09fbe441d08ba4d36c1fd810c. We reproduce the finite identity and its typed coupling map here to make the join complete; no novelty or limiting estimate is claimed.

All quantities in this section are at the original observation $\theta = 0$, with $N \geq q$. Write $A^\sharp = G_N^{-1}A^*G_N$ and $\mathcal{H}_N = A^\sharp + A - kI$, acting on the original (C, G_N) . Let C_+ be the full generalized spectral subspace with $\Re\lambda > c$, let $C_- = C_+^{\perp G_N}$, and let $P : C \rightarrow C$ be their orthogonal projection onto C_+ . Suppose first $C_+ \neq 0$. Its trace L from (TVB.35) is positive, so $\epsilon_N > 0$. Choose G_N -unit eigenvectors u_+, u_- of $\mathcal{H}_N$ at $\epsilon_N, -\epsilon_N$; selfadjointness makes them orthogonal. They specify only the coefficient injection

$$E : \mathbb{C}^2 \longrightarrow C, \quad E(z_+, z_-) = z_+ u_+ + z_- u_-, \quad E^\sharp E = I_2, \quad M = E^\sharp P E. \quad (\text{TVB.39})$$

Both M and $I_2 - M$ are positive semidefinite, since their quadratic forms are respectively the squared original norms of $\mathbf{P}Ez$ and $(1 - \mathbf{P})Ez$. In the original orthogonal decomposition define

$$B = A|_{C_+}, \quad D^{\text{op}} = (1 - \mathbf{P})A|_{C_-}, \quad \mathbf{C} = \mathbf{P}A|_{C_-} : C_- \longrightarrow C_+.$$

Invariance of C_+ gives $A = \begin{pmatrix} B & \mathbf{C} \\ 0 & D^{\text{op}} \end{pmatrix}$. Its adjoint has the conjugate-transpose blocks in the original restricted metrics, so $\mathbf{P}\mathcal{H}_N|_{C_-} = \mathbf{C}$ exactly.

Put $a_+ = \|\mathbf{P}u_+\|^2$, $a_- = \|\mathbf{P}u_-\|^2$, and $z = \langle \mathbf{P}u_+, \mathbf{P}u_- \rangle_{G_N}$. The compression M has these two diagonal entries and off-diagonal entries $z, \bar{z}$. The original rank-two spectral expansion gives $L = \epsilon_N(a_+ - a_-)$ and

$$\begin{aligned} \|\mathbf{C}\|_{\text{HS}, G_N}^2 &= \text{Tr}(\mathbf{P}\mathcal{H}_N^2\mathbf{P}) - \text{Tr}((\mathbf{P}\mathcal{H}_N\mathbf{P})^2) \\ &= \epsilon_N^2(a_+ + a_- - a_+^2 - a_-^2 + 2|z|^2). \end{aligned}$$

The cross trace in the second equation is $|\langle v, w \rangle|^2$ for the two rank-one maps $v\langle v, \cdot \rangle$ and $w\langle w, \cdot \rangle$, as follows by composing them. Subtract the displayed norm and $\epsilon_N^2(a_+ - a_-)^2$ from ϵ_N^2 ; direct expansion proves

$$\epsilon_N^2 - L^2 = \|\mathbf{C}\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}. \quad (\text{TVB.40})$$

Both determinants are nonnegative by the quadratic forms after (TVB.39). Adding (TVB.40) to (TVB.32) cancels precisely the intermediate ϵ_N^2 and gives all four nonnegative losses

$$\boxed{\alpha_{N+1} \sinh^2 s_N - L^2 = \alpha_{N+1}(e^{t_N} - \cosh s_N)^2 + \psi_N^2 + \|\mathbf{C}\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}.} \quad (\text{TVB.41})$$

Every adjoint, projection, norm and coefficient is for the same original minimum metric as the Toda volumes.

The spectral projector has an exact map to the coupling. Factor the minimal polynomial of A as m_+m_- over the selected and remaining full generalized blocks, with their original exponents. Bezout polynomials a, b with $am_+ + bm_- = 1$ give $Q = b(A)m_-(A)$: it is identity on each selected block and zero on each other block. Thus $Q^2 = Q$, $AQ = QA$, and in the preceding orthogonal decomposition $Q = \begin{pmatrix} I & Y \\ 0 & 0 \end{pmatrix}$ for a unique map $Y : C_- \rightarrow C_+$. Block multiplication proves

$$\mathcal{S} : \text{Hom}(C_-, C_+) \longrightarrow \text{Hom}(C_-, C_+), \quad \mathcal{S}(Y) = BY - YD^{\text{op}} = \mathbf{C}, \quad Q - \mathbf{P} = \begin{pmatrix} 0 & Y \\ 0 & 0 \end{pmatrix}. \quad (\text{TVB.42})$$

This morphism is invertible with its full original nilpotent parts:

$$\mathcal{S}^{-1}(Z) = \int_0^\infty e^{-tB} Z e^{tD^{\text{op}}} dt. \quad (\text{TVB.43})$$

Indeed every selected eigenvalue has real part greater than c and every remaining one has real part at most c . On a full block at λ the first exponential is exactly $e^{-t\lambda} \sum_{j < \ell_\lambda} (-t)^j (B - \lambda I)^j / j!$, with its corresponding projector, and the second has the same formula with positive t on its remaining blocks. Summing these finite expressions gives a polynomial times a strictly decaying exponential for their product in (TVB.43). This proves convergence in the original operator norm. Differentiating the integrand gives $-Be^{-tB}Ze^{tD^{\text{op}}} + e^{-tB}Ze^{tD^{\text{op}}}D^{\text{op}}$. Its integral has lower endpoint Z and zero upper endpoint, proving the inverse identity. If $BY - YD^{\text{op}} = 0$, the same product with $Z = Y$ is constant and decays to zero, hence $Y = 0$. This proves uniqueness as well. The space C_- is nonzero: otherwise $C_+ = C$ and $L = \text{Tr } \mathcal{H}_N = 0$, contradicting $L > 0$. Together with $C_+ \neq 0$ this makes the unit sphere in $\text{Hom}(C_-, C_+)$ nonempty. The actual smallest singular value $s_{B,D} > 0$ of $\mathcal{S}$ in the original Hilbert–Schmidt pairing exists by compactness of its unit sphere and injectivity. Consequently $\|\mathbf{C}\|_{\text{HS}, G_N} \geq s_{B,D}\|Q - \mathbf{P}\|_{\text{HS}, G_N}$. No bound on $s_{B,D}$ independent of the arithmetic inputs is used.

Retaining the phase and coupling in (TVB.41) gives the finite strengthening

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{\sqrt{L^2 + \psi_N^2 + \|\mathbf{C}\|_{\mathrm{HS}, G_N}^2}}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.44})$$

It follows by subtracting only the nonnegative imbalance and compression-determinant terms and applying the same inverse sinh as in (TVB.35). Summation over the disjoint indices $N = N_0 + 2j$ has the same exact endpoints as (TVB.36), with this degree-dependent square root in each summand. If $C_+ = 0$, use (TVB.32) directly; no positive-eigenline compression is asserted when $\epsilon_N = 0$. At $N = q - 1$, the scalar term instead uses the separate endpoint expression (TVB.38). These identities supply no asymptotic estimate and change no arithmetic class.

10 Crosswalk and exact continuing arithmetic quantity

CV.1–CV.3 are (TVB.22)–(TVB.24), with the additional proved identification $D_N^{\mathrm{CV}} = \mathfrak{B}_m / \mathfrak{D}_{N+1}$. CV.4–CV.5 become (TVB.27)–(TVB.29), where the whole complex cross term is $i\omega_N(\sigma - \ell'_N)$. CV.6 is the minimum lift (TVB.14) and its determinant-line map after (TVB.15). CV.10–CV.10a are (TVB.30)–(TVB.31), now with the exact two-loss identity (TVB.34). CV.11–CV.12 are precisely (TVB.37)–(TVB.38). The new operator refinement is (TVB.20)–(TVB.21): it computes both individual rank-one curvature costs, including their shared retained quantity $\alpha_{N+1}\delta_N$.

For an actual quartet polynomial the exterior multiplicities and their lower trace L remain those of that exact polynomial; for a larger packet they remain a lower contribution to its actual full trace. All admission thresholds here use its actual $q = \deg \chi$. The finite formulas prove an upper estimate by the exact source and relation determinants and prove the telescope (TVB.36). They do not estimate the growth of α_{N+1} or the joint determinant ratio as k and N grow. That continuing arithmetic question is the original expression (TVB.27), equivalently the positive-minor source norm in (TVB.28)–(TVB.31), with both losses (TVB.32) available without changing any of its inputs.